GANGA INSTITUTE OF TECHNOLOGY AND MANAGEMENT
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A PROJECT REPORT ON

“SMART TEMPERATURE CONTROL SYSTEM

SUBMITTED OF THE REQUIREMENTS FOR TASK - 3

MAINCRAFT TECHNOLOGY INTERNSHIP :

INTRODUCTION TO EMBEDDED SYSTEM & IOT DEVICES DESIGN

SUBMITTED TO :

MAINCRAFT TECHNOLOGY

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PROGRAM- INTRODUCTION TO EMBEDDED SYSTEM & IOT DEVICE DESIGN

SMART TEMPERATURE CONTROL SYSTEM

INTRODUCTION TO EMBEDDED SYSTEM

What is an Embedded System

An Embedded System is a specialized computer system that combines hardware and software to perform a specific task efficiently. Unlike a general-purpose computer, an embedded system is designed to carry out one dedicated function with high reliability and low power consumption.

Introduction

The Smart Temperature Monitoring System using ESP32 and DHT22 is an IoT-enabled embedded system designed to monitor environmental temperature and humidity in real time and automatically generate alerts based on predefined temperature thresholds. The system utilizes the DHT22 digital sensor to accurately measure temperature and humidity, while the ESP32 development board processes the sensor data and executes the control logic

Objective

The main objective of the Smart Temperature Monitoring System using ESP32 and DHT22 is to develop an intelligent embedded system capable of continuously monitoring environmental temperature and humidity and automatically generating alerts when the temperature exceeds a predefined threshold. The project aims to interface the DHT22 sensor with the ESP32 development board, process real-time sensor data, and implement automation using IF–ELSE decision-making logic.

Project Overview

The Smart Temperature Monitoring System using ESP32 and DHT22 continuously monitors the surrounding temperature and humidity using the DHT22 sensor. The sensor sends digital data to the ESP32, which processes the readings and compares the temperature with a predefined threshold value. If the temperature rises above 30°C, the ESP32 automatically turns ON the LED and buzzer to generate visual and audible alerts.

Working Principle

The Smart Temperature Monitoring System using ESP32 and DHT22 operates on the principle of real-time sensing, digital data processing, and automated control. The DHT22 sensor continuously measures the surrounding temperature and humidity and transmits the digital data to the ESP32 microcontroller. The ESP32 processes the received data and compares the measured temperature with a predefined threshold value using IF–ELSE decision-making logic.If the measured temperature exceeds the preset threshold (e.g., 30°C), the ESP32 automatically activates the LED and buzzer to provide visual and audible alerts. When the temperature falls below the threshold, the ESP32 deactivates both the LED and buzzer. Simultaneously, the temperature and humidity readings are displayed on the Serial Monitor for continuous real-time monitoring.

Component Required

- **ESP32 Development Board**
- **DHT22 Temperature & humidity**
- **Buzzer**
- **LED**
- **220Ω Resistor**
- **Breadboard**
- **Jumper Wires**

Component Description

- **ESP32 Development Board -** The ESP32 is the main controller of the system. It receives temperature data from the TMP36 sensor, processes the information, and controls the output device based on the programmed automation logic\

- **DHT22 Temperature & humidity -** The TMP36 is a precision analog temperature sensor that measures the surrounding temperature and generates an output voltage proportional to the measured temperature.

- **Active Buzze**r - The buzzer produces an audible alarm when the temperature exceeds the threshold, alerting the user.

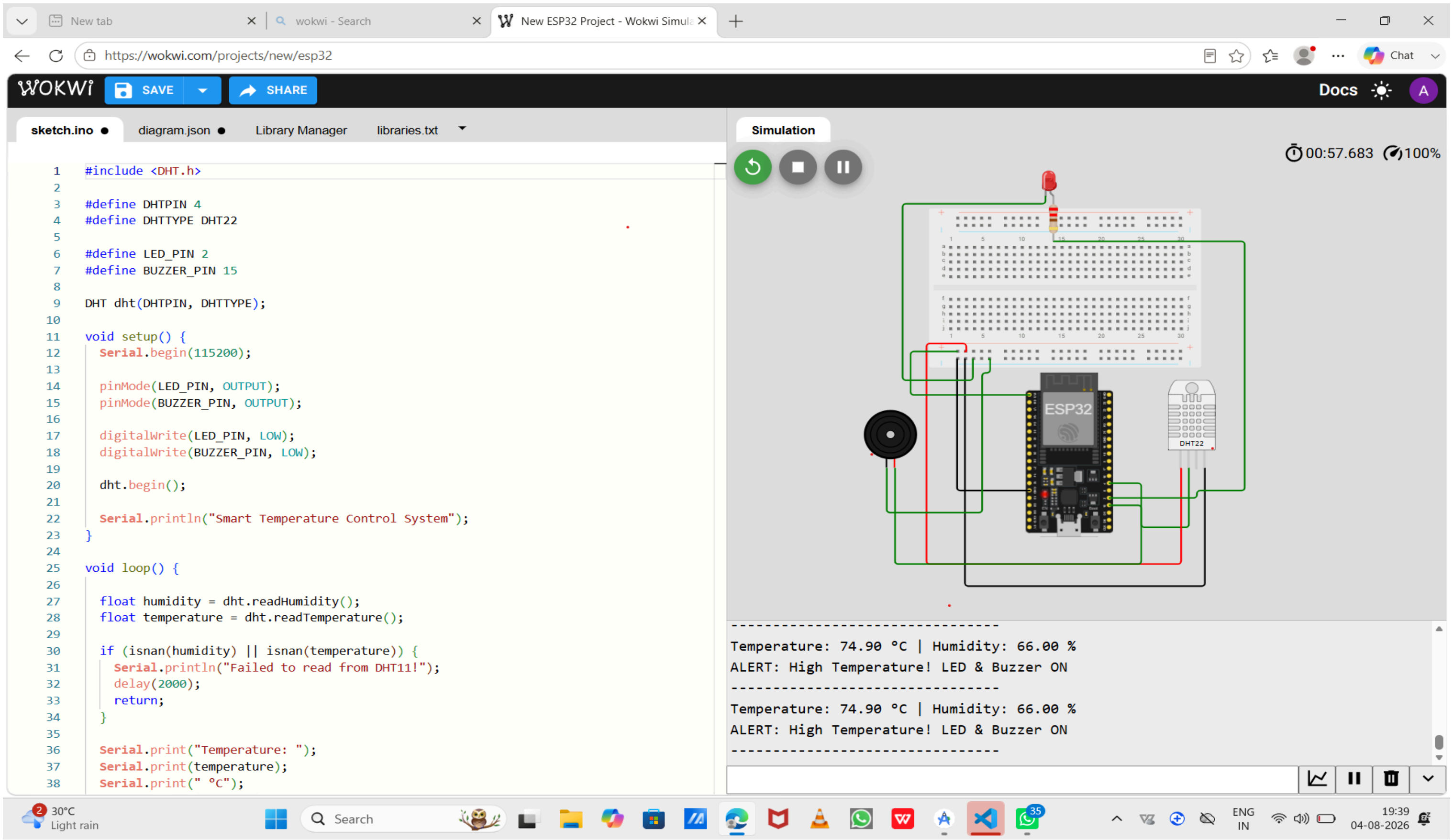
- **LED -** The LED (Light Emitting Diode) serves as a visual indicator. It turns ON when the temperature exceeds the predefined threshold and turns OFF when the temperature returns to the normal range.

- **220 Ω Resistor-** The 220 Ω resistor is connected in series with the LED to limit the current flow. It protects the LED from excessive current and ensures safe operation.

- **Breadboard -** The breadboard is used to assemble the circuit without soldering. It allows easy modification, testing, and debugging of the hardware connections.

- **Jumper Wires -** Jumper wires are used to establish electrical connections between the ESP32, TMP36 sensor, LED, resistor, and breadboard, ensuring reliable signal and power transmission.

Circuit Diagram



Automation Logic

The ESP32 continuously receives temperature and humidity data from the DHT22 sensor. The measured temperature is compared with a predefined threshold value (30°C) using IF–ELSE decision-making logic.

- ❖ If the temperature is greater than 30°C, the ESP32 automatically turns ON the LED and activates the buzzer to indicate a high-temperature condition.
- ❖ If the temperature is 30°C or below, the ESP32 turns OFF the LED and buzzer, indicating that the temperature is within the normal range.
- ❖ The current temperature and humidity values are simultaneously displayed on the Serial Monitor for real-time monitoring.

Programming

```
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT22

#define LED_PIN 2
#define BUZZER_PIN 15

DHT dht(DHTPIN, DHTTYPE);

void setup() {
 Serial.begin(115200);

 pinMode(LED_PIN, OUTPUT);
 pinMode(BUZZER_PIN, OUTPUT);

 digitalWrite(LED_PIN, LOW);
 digitalWrite(BUZZER_PIN, LOW);

 dht.begin();

 Serial.println("Smart Temperature Control System");
}

void loop() {

 float humidity = dht.readHumidity();
 float temperature = dht.readTemperature();

 if (isnan(humidity) || isnan(temperature)) {
  Serial.println("Failed to read from DHT11!");
```

```
    delay(2000);
    return;
  }

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C");

  Serial.print(" | Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  if (temperature > 30) {
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);

    Serial.println("ALERT: High Temperature! LED & Buzzer ON");
  }
  else {
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    Serial.println("Temperature Normal. LED & Buzzer OFF");
  }

  Serial.println("--------------------------------");
  delay(2000);
}
```

Advantages

- **Real-Time Monitoring :** Continuously monitors temperature and humidity using the DHT22 sensor.

- **Automatic Alert System:** Automatically activates the LED and buzzer when the temperature exceeds the preset threshold.

- **High Accuracy:** The DHT22 sensor provides more accurate temperature and humidity readings than basic sensors.

- **Low Power Consumption:** The ESP32 is energy-efficient, making the system suitable for continuous operation.

- **Cost-Effective:** The project uses affordable and easily available components.

Real world Application

- **Smart Home Automation** – Monitors indoor temperature and automatically triggers alerts when the temperature exceeds a safe limit.

- **Industrial Temperature Monitoring** – Continuously monitors temperature in factories and production units to prevent overheating of equipment.

- **Server Rooms and Data Centers** – Detects high temperatures and provides timely alerts to protect servers and networking devices.

- **Greenhouses and Agriculture** – Monitors environmental conditions to maintain optimal temperature for healthy plant growth.